		$\frac{I_s}{} = \frac{N_p}{} = \frac{}{200} \times 0.60 = 4.0$ A
28	**A**	The square of the wave function is proportional to the probability of finding the electron. Options B, C and D have positions where the square of the wave function is greatest. There is a high probability of finding the electron at one position, hence a low uncertainty in position. From Heisenberg's Uncertainty Principle, the uncertainty in momentum is hence large. Option A has many positions where the electron could be found with equal probability so there is a large uncertainty in position. This implies that the uncertainty in momentum is smallest.
29	**A**	For an element to be present in the nebula, all the spectral lines of the element must